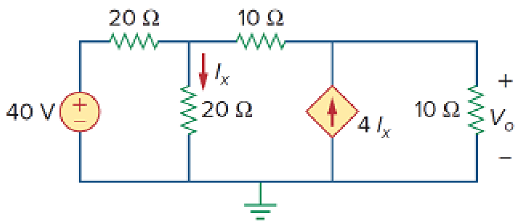
Chapter 3, Problem 12P

Problem

Using nodal analysis, determine *Vo* in the circuit in Fig. 3.61.

Figure 3.61:



Step-by-step solution

Step 1 of 4

Draw the following circuit diagram shown in Figure 1.

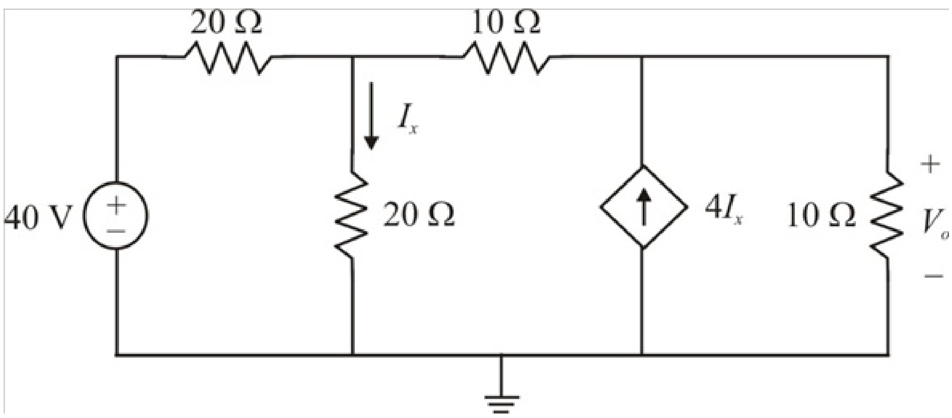


Figure 1

Step 2 of 4

The circuit with nodes and currents labeled is shown in Figure 2.

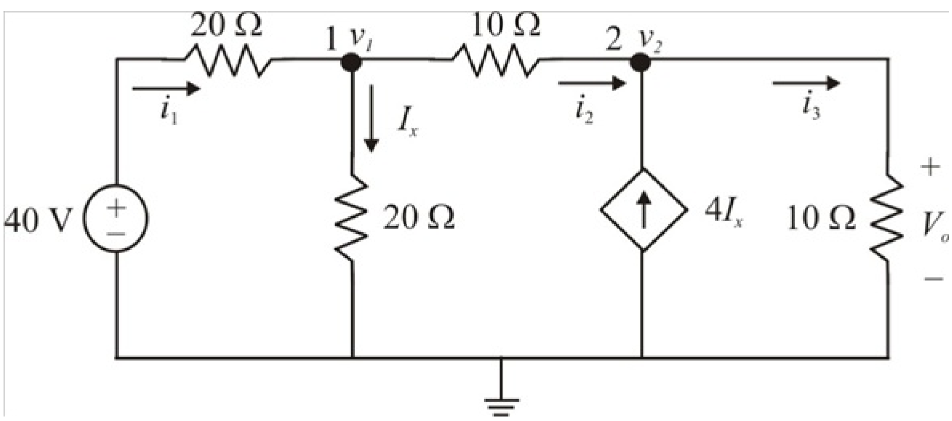


Figure 2

Step 3 of 4

Applying Kirchhoff's current law, the sum of currents entering the node is equal to the sum of the currents leaving from that node.

At the node 1, Kirchhoff's current law gives,

$$i_1 = I_x + i_2$$
$$\frac{40 - v_1}{20} = \frac{v_1}{20} + \frac{v_1 - v_2}{10}$$

Multiplying by 20 on both sides, we get

$$40 - v_1 = v_1 + 2v_1 - 2v_2$$
$$4v_1 - 2v_2 = 40$$

Therefore $4v_1 - 2v_2 = 40$ …… (1)

At the node 2, Kirchhoff's current law gives,

$$i_2 + 4I_x = i_3$$
$$\frac{v_1 - v_2}{10} + 4\frac{v_1}{20} = \frac{v_2}{10}$$

Multiplying by 10 on both sides, we get

$$v_1 - v_2 + 2v_1 = v_2$$
$$3v_1 - 2v_2 = 0$$

Therefore $3v_1 - 2v_2 = 0$ …… (2)

Step 4 of 4

Subtract equation (2) from equation (1), we get

$$\begin{array}{r} 4v_1 - 2v_2 = 40 \\ 3v_1 - 2v_2 = 0 \\ \hline v_1 = 40\,\text{V} \end{array}$$

Substituting the value of v_1 in equation (1), we get

$$4v_1 - 2v_2 = 40$$
$$4(40) - 2v_2 = 40$$
$$2v_2 = 160 - 40$$
$$v_2 = 60\,\text{V}$$

Since the current source is parallel to the 10 ? resistor, the voltage v_2 is equal to the voltage across the 10 ? resistor, V_o

$$V_o = v_2$$
$$= 60\,\text{V}$$

Therefore the voltage V_o is $\boxed{60\,\text{V}}$.